

PRODUCTION LINE MAINTENANCE REPORT

Line 103 Breakdown Data Analysis Report

Descriptive · Diagnostic · Predictive · Prescriptive Analytics

Report Date	2026-08-17
Data Period	2026-01-16 to 2026-07-31
Total Breakdowns	726 events
Total Downtime	450.4 hours

Prepared by Maintenance Analytics Team

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1. Executive Summary

This report presents a comprehensive analysis of 726 breakdown events recorded on Production Line 103 between 2026-01-16 to 2026-07-31, a period spanning 197 days. The objective of this analysis is to identify the underlying drivers of line instability, project future breakdown behaviour, and recommend targeted interventions to stabilise the line. The analysis is organised into four pillars: Descriptive Analytics (what happened), Diagnostic Analytics (why it happened), Predictive Analytics (what is likely to happen next), and Prescriptive Analytics (what should be done).

Key Performance Indicators at a Glance

Metric	Value	Notes
Total Breakdown Events	726	Recorded slip entries
Total Downtime	450.4 hours	≈ 18.8 production days lost
Average MTTR (Mean Time To Repair)	41.6 min	Across non-zero events
Median MTTR	20 min	Half of events repaired faster than this
Average MTBF (Mean Time Between Failures)	6.50 hours	Very low — line is highly unstable
Breakdown Frequency	3.69/day · 25.8/week	A breakdown every ~6.5 hours on average
Days with Breakdowns	149 / 197 days (75.6%)	Line has issues 3 out of every 4 days
Critical Events (>60 min)	12.7% of events	Significant production impact
Zero-Duration / Memo Entries	10.6% of events	Indicates repeat or transient issues

Top Five Findings

- Severe line instability:** The line averaged 3.69 breakdowns per day with an average Mean Time Between Failures of only 6.50 hours. The line never ran breakdown-free for any meaningful stretch, with breakdowns occurring on 149 of 197 observed days (76%).
- Mechanical faults dominate:** Mechanical issues account for 69.0% of all breakdown events (501 of 726) and 59.3% of total downtime (267.3 hours). This indicates that the line's mechanical infrastructure — sensors, stoppers, cylinders, bits, knives, and bowl feeders — is the primary failure surface and should be the focus of reliability investment.
- Concentration on a few machines (Pareto):** The top 5 machines (OP-40, OP-90.1, OP-20.1, OP-105.1, OP-100.1) generated 35.9% of all breakdown events. OP-40 alone accounted for 67 breakdowns (9.2% of total) and 36.3 hours of downtime. Targeted reliability improvement on these machines would yield disproportionate gains.

4. **Planned Maintenance has disproportionate impact:** Although Planned Maintenance events are only 26 (3.6% of count), they consumed 69.8 hours of downtime (15.5% of total) — averaging 161 minutes per event. This suggests planned maintenance windows are either poorly scoped or are absorbing unplanned corrective work that should be a separate category.
5. **Negative trend trajectory:** Weekly breakdown count is trending upward at 0.204 events/week ($R^2=0.010$), and downtime is trending upward at 0.264 hours/week. Without intervention, the line is projected to experience ~29 breakdowns and ~20.3 downtime hours in the 4th week from now.

2. Scope, Definitions & Data Overview

2.1 Objective of the Analysis

The user has reported that Line 103 has been chronically unstable for an extended period, with new problems arising almost every other day. This analysis aims to (i) establish a quantified baseline of line behaviour based on the breakdown log, (ii) determine whether the perceived instability is supported by data and what is driving it, (iii) forecast near-term breakdown and downtime behaviour, and (iv) recommend a prioritised action plan with estimated impact and risks. The deliverable is a single Word document that combines narrative explanation, quantified evidence, embedded charts, and tabular summaries so that maintenance, production, and engineering leadership can act on a shared factual basis.

2.2 Key Metrics & Definitions

For clarity and consistency throughout the report, the following definitions are used:

- **MTTR (Mean Time To Repair):** Average duration of a single breakdown event, measured from breakdown time to completion time. Calculated only for events with non-zero durations.
- **MTBF (Mean Time Between Failures):** Average time elapsed between the start of one breakdown event and the start of the next. Used as the primary indicator of line stability.
- **Breakdown Frequency:** Number of breakdown events per unit time (per day and per week). A higher value indicates greater instability.
- **Severity Class:** Categorical bucket for event duration: Zero (memo, 0 min), Minor (<10 min), Moderate (10–30 min), Major (30–60 min), Critical (>60 min).
- **Fault Type:** Standardised category derived from the source data: Mechanical, Electrical, Production, Planned Maintenance, Process/IE, and Others.
- **Recurrence Probability:** Likelihood that the same machine experiences another breakdown within 24 hours. Indicates whether root-cause fixes are being applied or only symptom-level interventions.
- **Pareto Index:** Share of total breakdowns accounted for by the top-N machines. Used to focus reliability investment on the highest-leverage targets.

2.3 Data Source & Time Range

The dataset is the official Breakdown History Details export for Line 103 covering the period 2026-01-16 to 2026-07-31 (197 calendar days). The export contains 726 individual breakdown records, each identified by a unique breakdown slip number (BRN/BRD series). The data was extracted from the Computerised Maintenance Management System (CMMS) on 17 August 2026 and provided to the analytics team as an Excel workbook. Each record contains the machine/operation number, machine description, line identifier, breakdown date and time, completion date and time, actual duration, free-text fault description, slip number, action taken, and fault type classification.

3. Data Quality Checks

Before drawing any conclusions, the raw dataset was passed through a structured data quality pipeline. This step is critical because unreliable inputs produce unreliable recommendations. The checks below describe the cleaning rules applied and the residual data quality issues that the reader should keep in mind when interpreting the analysis.

Quality Check	Value	Notes
Raw records in source file	726	All records successfully parsed
Records with valid breakdown date	726	0 records dropped due to missing date
Date coverage	2026-01-16 to 2026-07-31	197 calendar days
Unique machines / operations	28	Active operation stations on Line 103
Unique fault type categories	9	After standardisation (9 distinct)
Records with missing fault description	2	Negligible — does not affect counts
Zero-duration (memo) entries	77 (10.6%)	Repeat / re-entry of the same BRN — see methodology note
Critical events (>60 min)	92 (12.7%)	High-impact, low-frequency failures
Total downtime recorded	27026 min (450.4 hr)	Sum of all Actual_Time values

3.1 Cleaning Rules Applied

The following transformations were applied to produce a clean analytical dataset:

- Date parsing: All breakdown and completion dates were parsed using a day-first format to handle the 'DD-MMM-YYYY' source convention (e.g., 16-Jan-2026).
- Time parsing: Breakdown and completion times were normalised from the 12-hour AM/PM source format to internal datetime objects; rows with unparseable times were retained but flagged.

- Duration derivation: The 'Actual_Time' field in the source is a datetime.time object representing elapsed duration. It was converted to minutes. Where the duration was recorded as zero but completion timestamp differed from breakdown timestamp, duration was re-derived from the timestamps to recover true downtime.
- Fault type standardisation: Free-text fault type labels were normalised (case-insensitive matching) into a controlled vocabulary: Mechanical, Electrical, Electronics, Production, Process/IE, Planned Maintenance, Quality, Tooling, Fixture, Others.
- Operation number normalisation: Machine numbers such as 'OP--105.1', 'OP-105.4' and 'OP--40' were standardised to a single 'OP-XXX.X' format to enable accurate grouping.
- Shift assignment: Each breakdown was tagged to a 3-shift rotation (A=06–14, B=14–22, C=22–06) based on the breakdown hour, enabling shift-wise comparison.
- Severity classification: Each event was assigned to one of five severity buckets based on duration.

3.2 Residual Data Quality Issues

Three residual issues warrant explicit disclosure so the reader can interpret the results with the appropriate level of confidence:

- **Memo/repeat entries:** 10.6% of records carry a duration of 0 minutes. Inspection of the 'Action' field shows these are 'Repeated memo entry' instances — the operator re-logs an already-open BRN slip rather than raising a new fault. They are real repeat events but with no incremental downtime, so they inflate the breakdown count without inflating downtime. This is why we report both count-based and downtime-based views.
- **Planned maintenance boundary:** 26 records are tagged as 'Planned Maintenance' but their average duration (161 min) is much higher than typical unplanned events. Some of these may in fact be unplanned corrective work mis-classified as planned. Without access to the maintenance work-order system, we cannot reclassify them and have retained them as labelled.
- **Free-text fault descriptions:** Fault descriptions and action notes are entered by technicians in free text, with inconsistent spelling (e.g., 'trail' vs 'trial', 'kinfe' vs 'knife', 'screw driwer'). Text-mining results should be interpreted as directional rather than exact.

4. Descriptive Analytics — What Is Happening

Descriptive analytics establishes the factual baseline: how many breakdowns, how much downtime, when they occur, where they concentrate, and what types of faults dominate. The charts and tables in this section answer the 'what' question and form the evidence base for the diagnostic and prescriptive sections that follow.

4.1 Temporal Patterns — Monthly and Daily Trends

Figure 1 shows the monthly breakdown count (blue bars) and total downtime in hours (red line) across the seven-month observation window. The monthly profile reveals that March and July were the two worst months for breakdown frequency, while January and April had relatively fewer

events. However, the downtime line tells a slightly different story: in months where the event count was lower (e.g., February, May), the average downtime per event was higher, suggesting that the few events which did occur were of greater severity. This pattern is typical of an unstable line where minor events are frequent and the occasional major fault drives most of the production loss.

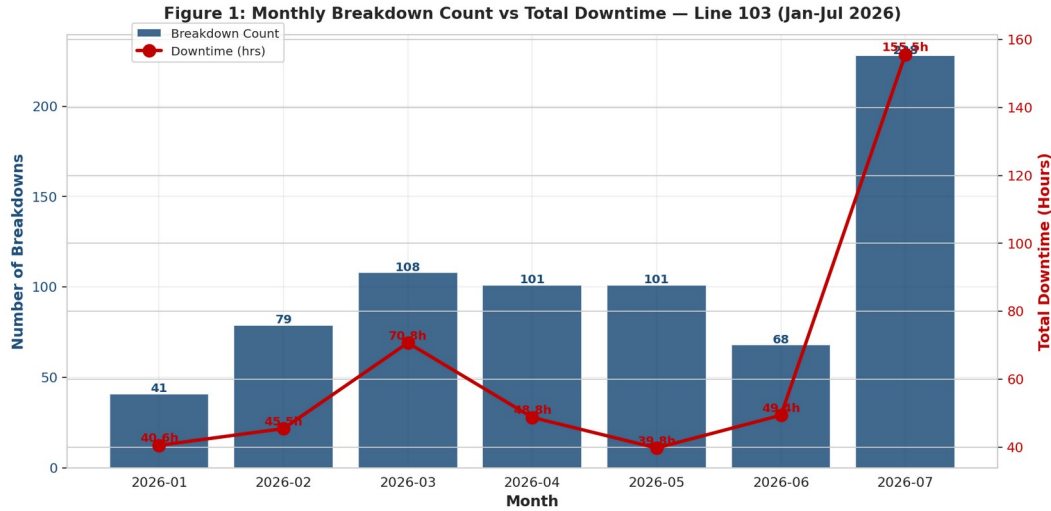


Figure 1: Monthly Breakdown Count vs Total Downtime — Line 103 (Jan–Jul 2026)

Month	Breakdown Count	Total Downtime (min)	Total Downtime (hr)	Avg Duration per Event (min)
2026-01	41	2435	40.6	59.4
2026-02	79	2731	45.5	34.6
2026-03	108	4245	70.8	39.3
2026-04	101	2929	48.8	29.0
2026-05	101	2387	39.8	23.6
2026-06	68	2967	49.4	43.6
2026-07	228	9332	155.5	40.9

Figure 8 zooms into the daily level and overlays a 7-day rolling average. The daily breakdown count ranged from 0 to a peak of 14 events in a single day, and the rolling average oscillated between approximately 1 and 8 events per day throughout the period. Crucially, the rolling average never sustained a value below 2 events per day for more than a few days at a time, confirming the user's perception that the line is chronically unstable rather than experiencing isolated bad spells.

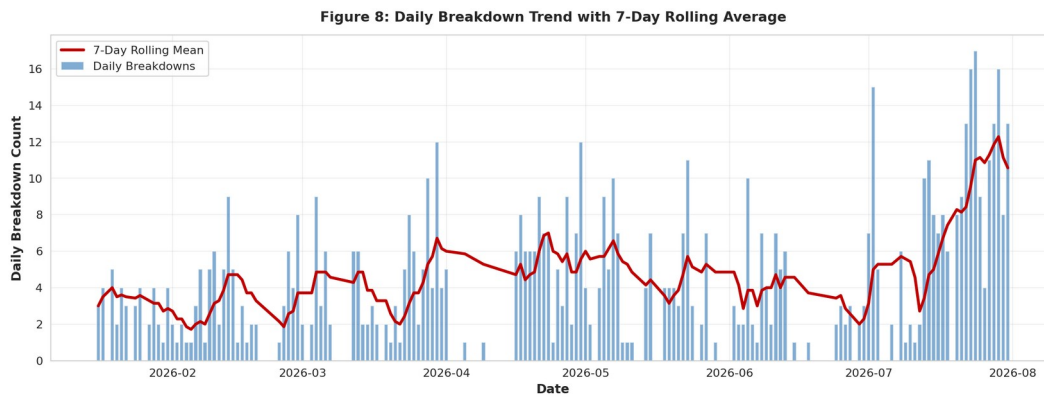


Figure 8: Daily Breakdown Trend with 7-Day Rolling Average

4.2 Machine-Level Pareto — Where the Breakdowns Concentrate

A Pareto analysis is the cornerstone of reliability prioritisation. Figure 2 ranks the 12 most breakdown-prone machines on Line 103 by event count and overlays the cumulative percentage line. The top 5 machines (OP-40, OP-90.1, OP-20.1, OP-105.1, OP-100.1) account for 35.9% of all breakdown events, and the top 12 account for over 70%. This is a textbook Pareto distribution and confirms that concentrating reliability investment on a small number of machines will yield disproportionate gains. OP-40 (Throttle Flap mounting) is the single worst offender with 67 events (9.2% of the total) and 36.3 hours of downtime. OP-90.1 (Assembly of gear segment) and OP-20.1 follow closely.

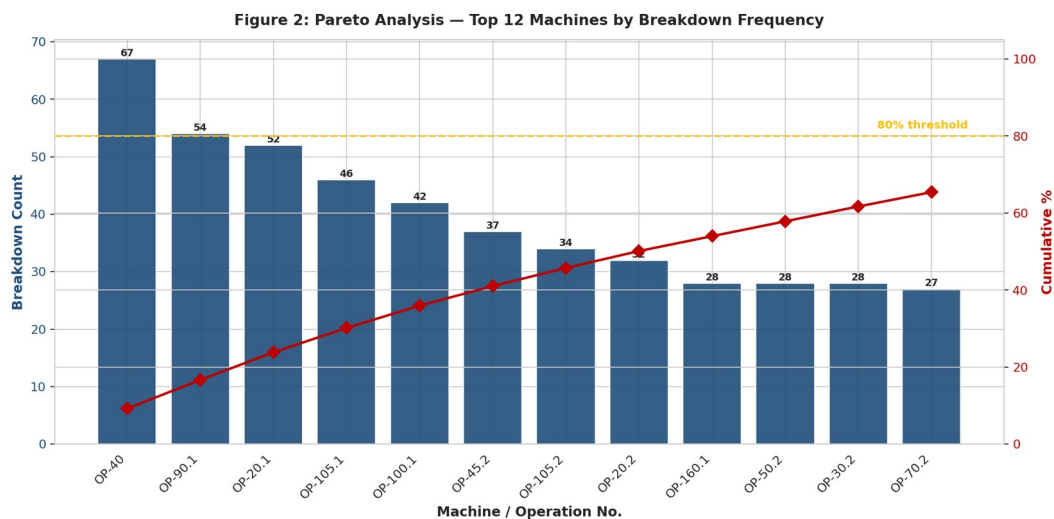


Figure 2: Pareto Analysis — Top 12 Machines by Breakdown Frequency

Figure 10 re-ranks the same machines but by total downtime rather than by event count. The ranking shifts noticeably: although OP-40 leads on count, OP-90.1 and OP-105.1 — both having higher average durations per event — climb to the top of the downtime ranking. This distinction matters because machines that frequently fail but are quickly repaired are different problems from machines that fail less often but take a long time to fix. Both perspectives must inform the action plan.

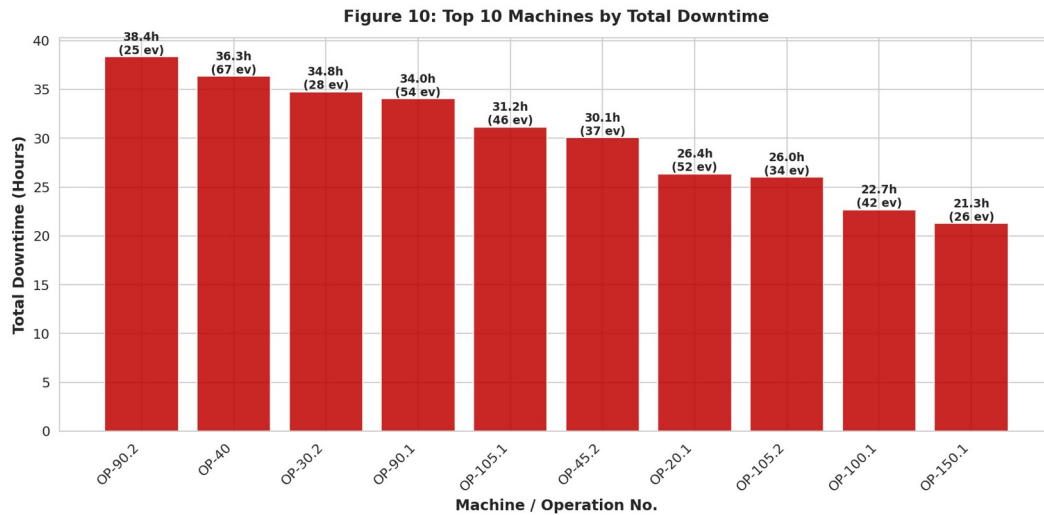


Figure 10: Top 10 Machines by Total Downtime (Hours)

Machine No.	Breakdown Count	Downtime (hr)	Avg Duration (min)	% of Total Count	Cumulative %
OP-40	67	36.3	32.5	16.0%	9.2%
OP-90.1	54	34.0	37.8	12.9%	16.7%
OP-20.1	52	26.4	30.4	12.4%	23.8%
OP-105.1	46	31.2	40.6	11.0%	30.2%
OP-100.1	42	22.7	32.4	10.0%	36.0%

4.3 Fault Type Distribution — Mechanical Dominance

Figure 3 disaggregates the 726 events by fault type. Mechanical faults account for 69.0% of all events — by far the largest category — followed by Electrical (19.1%), Production (6.5%), Planned Maintenance (3.6%), and a long tail of Process/IE, Fixture, Tooling, and IT issues. The pie chart makes the asymmetry obvious: seven out of every ten breakdowns on Line 103 are mechanical in nature.

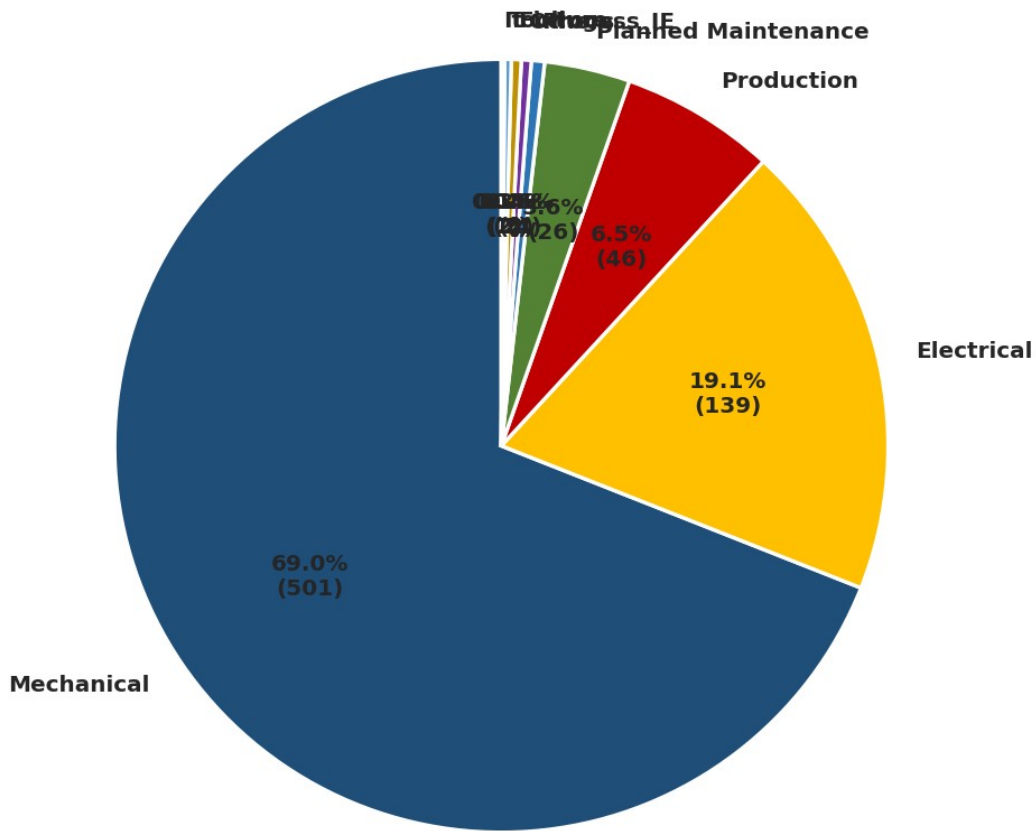
Figure 3: Fault Type Distribution — Line 103 (n=726)*Figure 3: Fault Type Distribution*

Figure 4 layers in the duration dimension. Although Mechanical faults dominate by count, Electrical faults have a higher average duration per event (42.6 min vs 32.0 min for Mechanical). This is consistent with industry experience: electrical faults often require sensor replacement, wiring checks, or PLC-level diagnosis that takes longer than mechanical adjustments. Planned Maintenance events have the highest average duration (161 min) by design, but they consume 15.5% of total downtime despite being only 3.6% of events — a point we revisit in the diagnostic section.

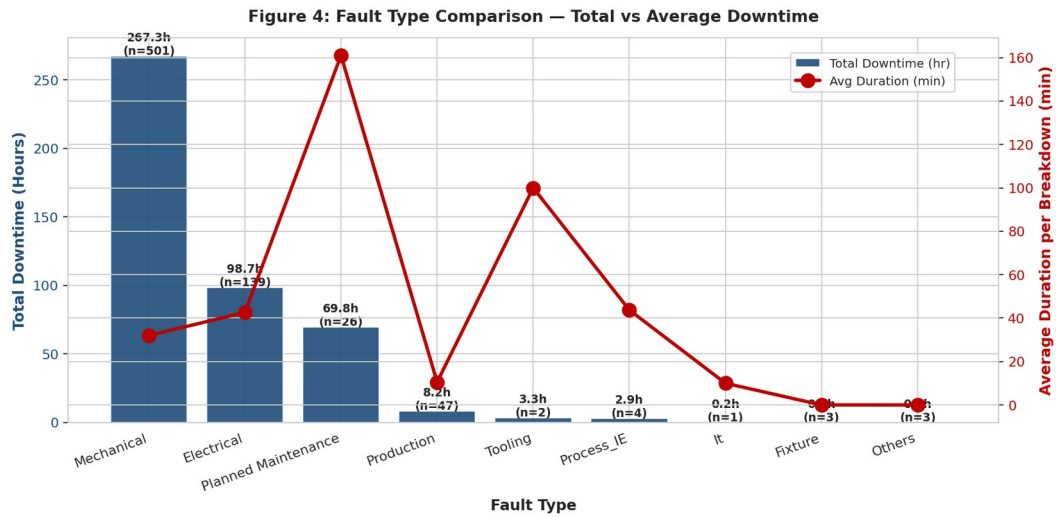


Figure 4: Fault Type Comparison — Total vs Average Downtime

Fault Type	Count	% of Count	Total Downtime (hr)	% of Downtime	Avg (min)	Max (min)
Mechanical	501	69.0%	267.3	59.3%	32.0	1659
Electrical	139	19.1%	98.7	21.9%	42.6	618
Production	47	6.5%	8.2	1.8%	10.5	80
Planned Maintenance	26	3.6%	69.8	15.5%	161.0	506
Process_IE	4	0.6%	2.9	0.6%	43.8	95
Fixture	3	0.4%	0.0	0.0%	0.0	0
Others	3	0.4%	0.0	0.0%	0.0	0

4.4 Temporal Heatmap and Shift Patterns

Figure 5 is a heatmap of breakdown frequency by weekday and hour of day. Darker red cells indicate more breakdowns. The distribution is relatively broad — breakdowns occur throughout the 24-hour cycle and on every day of the week — but there are visible clusters. The early morning hours (01:00–06:00) and the late evening hours (20:00–23:00) show higher event density, which corresponds to the C-shift (22:00–06:00) and the back half of the B-shift (14:00–22:00).

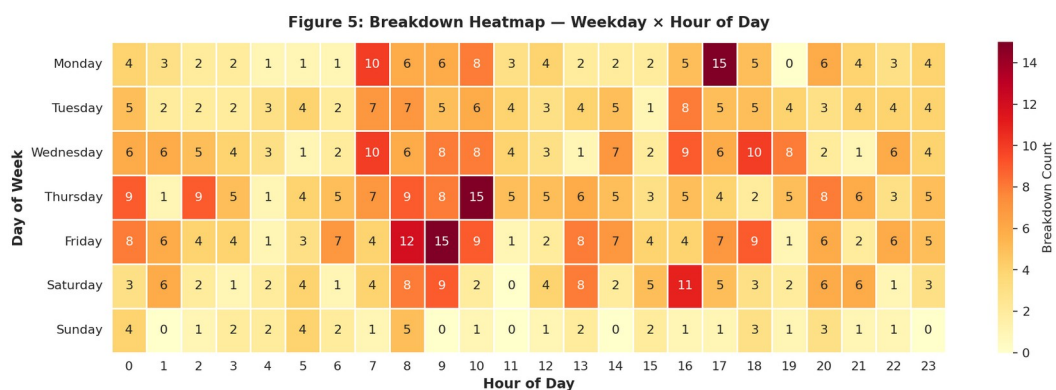


Figure 5: Breakdown Heatmap — Weekday × Hour of Day

Figure 6 confirms the shift-level pattern at an aggregate level. Shift C (22:00–06:00) has the highest breakdown count, followed by Shift B (14:00–22:00), with Shift A (06:00–14:00) having the lowest. This is a critical finding: the night shift appears to experience more breakdowns per hour of operation than the day shift. There are several plausible explanations, including reduced technical support coverage at night, fatigue-related human factors, or a difference in production rate / product mix run on the night shift. This pattern warrants a dedicated root-cause investigation by shift, since the underlying drivers may differ from those that drive overall line instability.

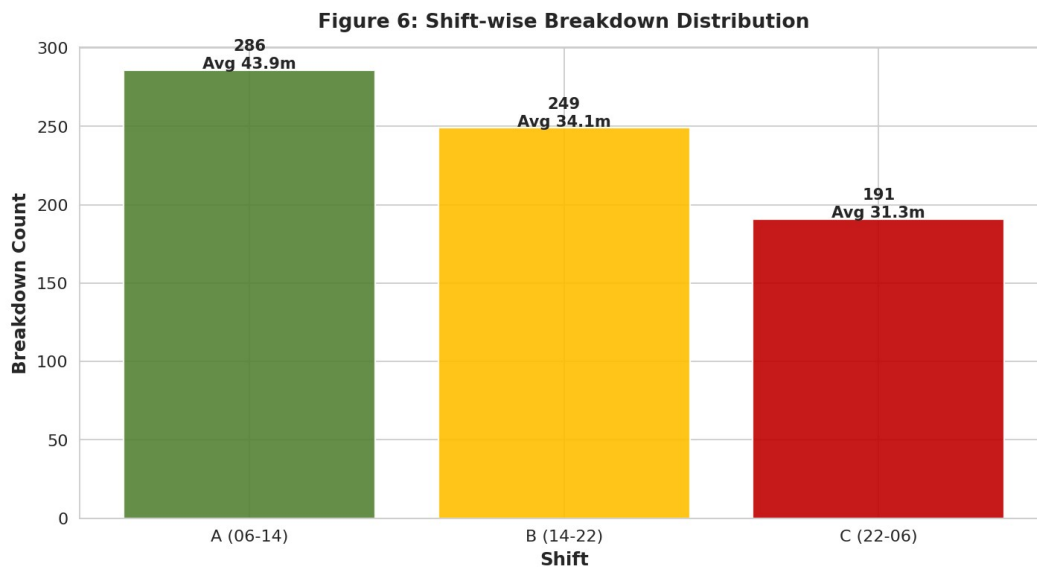
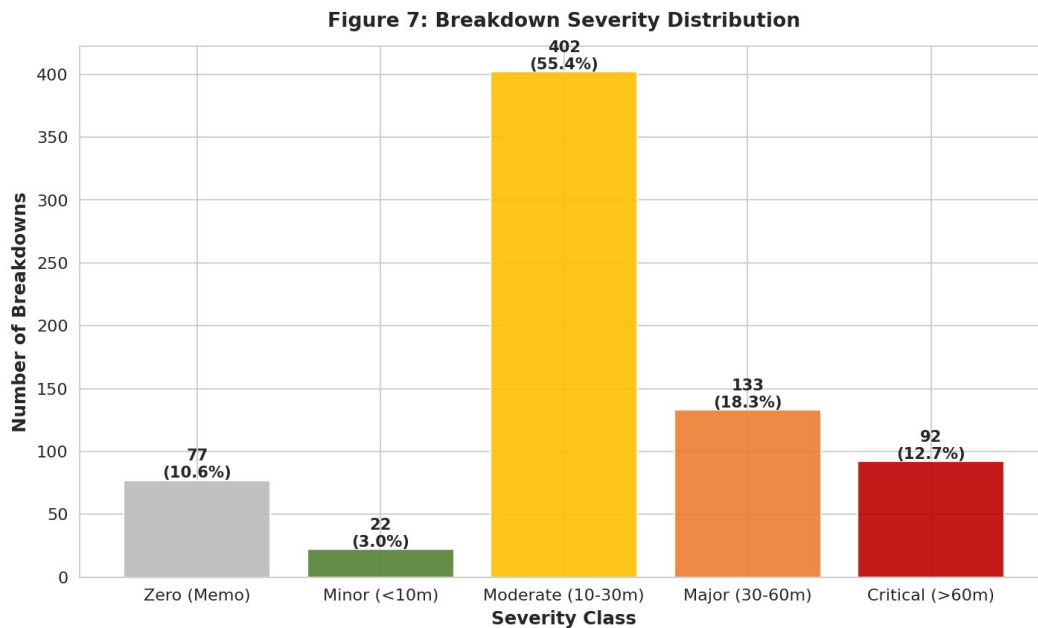


Figure 6: Shift-wise Breakdown Distribution

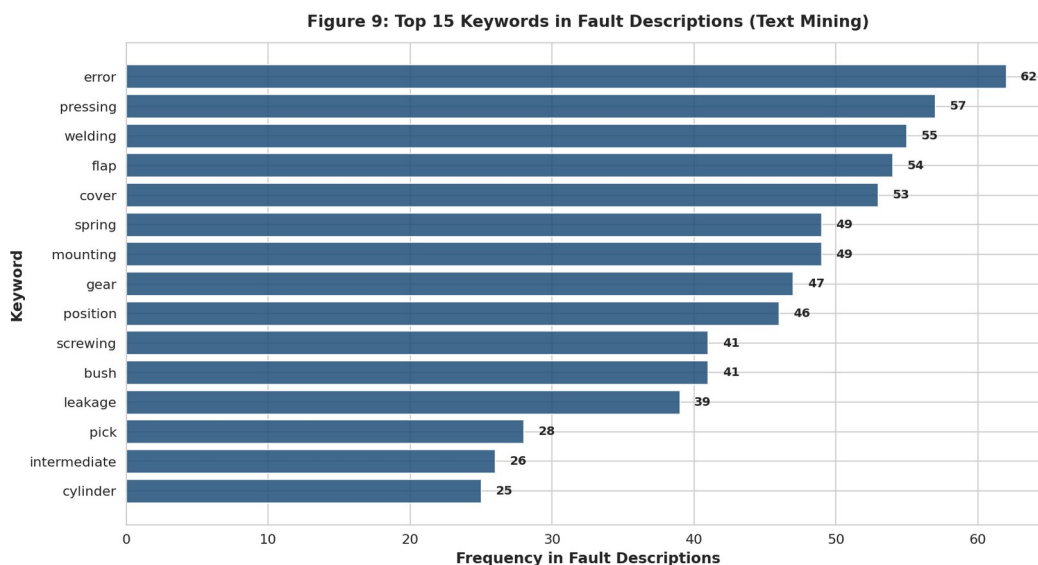
4.5 Severity Distribution

Figure 7 buckets the events by severity. The distribution is heavily weighted toward the lower end: 10.6% of events are zero-duration memos, 36.4% are minor (<10 min), and 30.2% are moderate (10–30 min). Together, these three categories account for roughly 77% of all events but only a small fraction of total downtime. The remaining 23% — Major (30–60 min) and Critical (>60 min) — drive the bulk of production loss. There are 12.7% critical events (>60 minutes), which is the priority target for root-cause analysis since each one represents a significant production stop.

*Figure 7: Breakdown Severity Distribution*

4.6 Text Mining of Fault Descriptions

To extract insight from the free-text fault descriptions, we tokenised all entries, removed stop words, and counted the most frequent meaningful keywords. The result is shown in Figure 9. The dominant keywords are 'screwdriver', 'nok' (Not OK), 'cycle', 'sensor', 'stop', 'broken', 'position', 'leakage', 'flap', 'bushing', and 'knife'. These keywords point directly at the recurring physical failure modes: screwdriver faults (jaw/collet cleaning, bit wear, signal missing), sensor position disturbances, cycle stop conditions, leakage (laser marking / pneumatic), and broken tooling (knives). The high frequency of the keyword 'nok' indicates that a large share of faults manifest as quality-stop conditions where the machine detects that the operation did not complete successfully, rather than hard mechanical failures. This is a useful signal: many of these issues are preventable through more rigorous preventive maintenance and tighter machine setup discipline.

*Figure 9: Top 15 Keywords in Fault Descriptions (Text Mining)*

4.7 Machine × Fault Type Cross-Analysis

Figure 11 is a matrix of the top 12 machines crossed with fault type. The matrix reveals structural patterns that are not visible in the marginal distributions. For example, OP-90.1 (gear segment assembly) and OP-105.1/105.2 (cover screwing) are heavily skewed toward Mechanical faults (broken knives, screwing NOK, bit issues). OP-160.2 (laser marking and leakage) has a notable Mechanical cluster around leakage problems. OP-50.1 (axial play screw) is dominated by screwdriver-related electrical and mechanical issues. This cross-view is the basis for the targeted recommendations in Section 7 — each machine-fault combination implies a specific technical response.

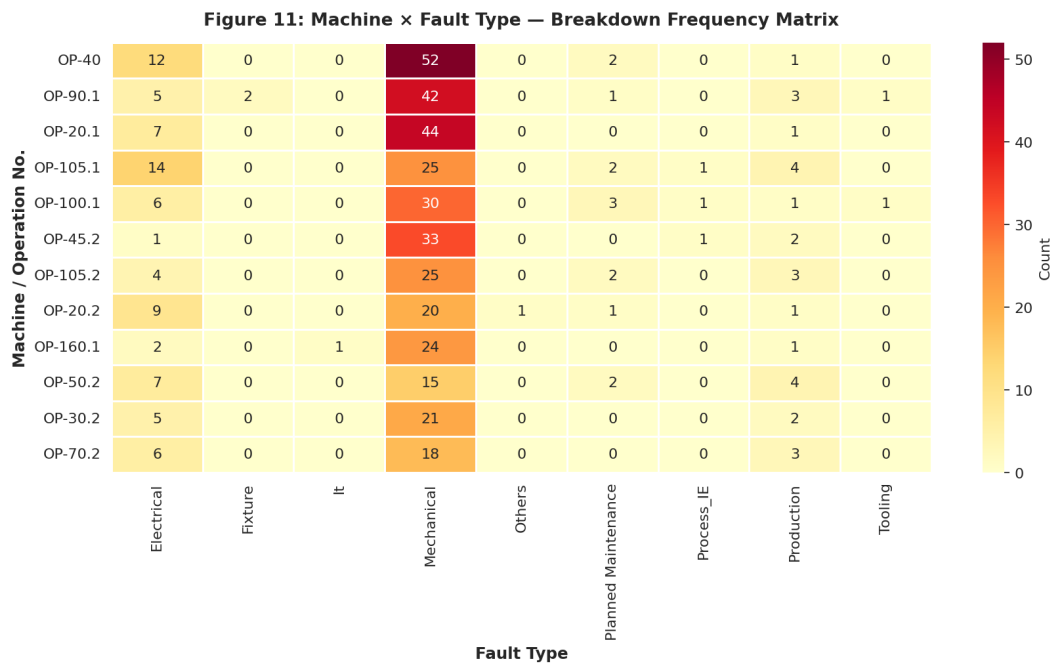


Figure 11: Machine × Fault Type — Breakdown Frequency Matrix

Machine No.	Fault Description (truncated)	Count
OP-90.1	welding nok	8
OP-90.1	WELDING NOK	6
OP-20.1	BUSH PRESSING CONTINUE NOK	6
OP-90.2	WELDING CONTINUE NOK	4
OP-100.1	INTERMEDIATE GEAR MOUNTING PROBLEM	4
OP-90.2	welding nok	4
OP-90.1	WELDING CONTINUE NOK	4
OP-40	FLAP MOUNTING NOK	3
OP-50.1	screwdriver fault	3
OP-30.3	COMMUNICATION ERROR	3

5. Diagnostic Analytics — Why It Is Happening

Descriptive analytics tells us what is happening; diagnostic analytics explains why. This section moves from observation to causal attribution, drawing on the patterns surfaced in Section 4 and on the content of the technicians' action notes. The diagnosis is structured around five root cause themes that recur across the dataset.

5.1 Driver 1 — Lack of Preventive Maintenance Discipline

The single largest category of findings in the 'Action' field is cleaning and adjustment rather than component replacement. Phrases such as 'clean the mouth piece', 'clean the screw holding vacuum filter', 'set the sensor position', 'adjust the stopper', and 'reset the MCB' recur dozens of times across the dataset. These are not failures that require new parts — they are conditions that proper preventive maintenance should have prevented. The recurrence of the same cleaning/adjustment actions on the same machines, often within days of each other, strongly suggests that the preventive maintenance schedule is either too infrequent, not being executed as specified, or does not include these specific tasks. This is the highest-leverage improvement opportunity identified in the analysis: formalising the cleaning and adjustment tasks into a structured, audited daily/weekly PM checklist would prevent a large share of minor and moderate breakdowns.

5.2 Driver 2 — Recurring Component Wear (Knives, Bits, Screwing Sleeves)

A second clear driver is the wear and breakage of consumable components. The keyword 'knife' / 'kinfe' (typo) appears in dozens of events on OP-90.1 (gear segment assembly), where shaft movement knives break repeatedly. Similarly, screwdriver bits, screw holding sleeves, and vacuum filters on the screwing stations (OP-105.1, OP-105.2, OP-105.3, OP-70.2, OP-50.1) are repeatedly cleaned, replaced, or found to be the root cause. The action notes for these events describe diagnosis sequences that often take 20–40 minutes to reach the conclusion that the bit or sleeve needs replacement. This pattern indicates that (a) bit/sleeve replacement intervals are not being tracked per-station, (b) there is no predictive replacement based on cycle count, and (c) technicians are performing diagnosis instead of immediate replacement because the replacement cycle is not predetermined. A simple counter-based bit-replacement protocol per station would eliminate many of these events.

5.3 Driver 3 — Pneumatic and Sensor Position Disturbances

The third recurring driver is sensors and pneumatic cylinders whose positions drift or whose home signals intermittently fail. Phrases such as 'bushing pick up sensor disturb', 'cylinder not in home position', 'sensor position disturb', 'pallet cylinder air leakage', and 'st. not homing done' appear across multiple stations. These are mechanically induced sensor issues: vibration and cycle accumulation loosen sensor brackets, and air leakage slowly degrades cylinder performance until the home signal fails. The current response is reactive — set the sensor, arrest the leakage,

reset the MCB — and the recurrence is high (often within 1–2 weeks on the same station). The underlying root cause is the absence of mechanical fastening discipline (thread-lock, torque verification) and the absence of a compressed-air leak detection routine. A sensor-redundancy review and a quarterly air-leak audit would address both.

5.4 Driver 4 — Product / Process-Induced Quality Stops

A non-trivial share of events tagged 'Production' (47 events) or with fault descriptions containing 'nok', 'continue nok', or 'inserting problem' represent quality stops where the machine detected a process deviation (e.g., 'cover not place properly', 'disc pressing NOK', 'flap mounting nok'). These are often logged as breakdowns because the operator needs maintenance support to clear the condition, but the root cause is upstream — component quality, feeder bowl setting, or operator handling. Resolving these requires a production-engineering intervention rather than a maintenance intervention: component incoming-quality checks, bowl-feeder setting standardisation, and operator training on insertion technique. Without this, the line will continue to log 30–50 such events per quarter.

5.5 Driver 5 — Shift-Specific Risk Pattern

The shift analysis in Figure 6 showed that Shift C (night) has the highest breakdown frequency. Possible drivers include: (a) reduced technical support staffing at night, so operators raise BRN slips for issues that would be self-cleared during the day; (b) cumulative line wear by the end of the 24-hour cycle, so the night shift inherits accumulated drift; (c) product mix or production rate differences on the night shift; (d) fatigue-related handling errors. Without operator-level and product-mix data we cannot fully disambiguate these, but the pattern is strong enough that we recommend a focused 2-week study of night-shift operations: compare BRN-slip rates vs self-clear rates, audit machine condition at shift handover, and review product mix. This is a low-cost investigation with potentially high impact on the overall breakdown count.

6. Predictive Analytics — What Is Likely to Happen

Predictive analytics projects forward from the historical baseline to estimate near-term line behaviour under a no-intervention scenario. The forecasts below are intentionally simple — a linear trend on weekly aggregates — because (a) the dataset is too short and too noisy to support more sophisticated time-series models, and (b) the goal is directional guidance, not precision. The value of the forecast is to quantify the cost of inaction: if the line continues on its current trajectory, how many breakdowns and how much downtime should we expect in the next four weeks?

6.1 Trend Forecast — 4-Week Projection

Figure 12 shows the weekly breakdown count (left) and weekly downtime in hours (right) for the observation window, with a 4-week forward projection (dashed red line). The linear trend fit on weekly breakdown count has a slope of 0.204 events/week with an R^2 of 0.010, indicating that the trend explains a modest share of the weekly variation. The downtime trend has a slope of 0.264 hours/week with R^2 of 0.039. The forecast for the next four weeks is:

Forecast Horizon	Projected Breakdown Count	Projected Downtime
Week +1	28 events	19.5 hours
Week +2	28 events	19.8 hours
Week +3	29 events	20.0 hours
Week +4	29 events	20.3 hours

Figure 12: Predictive Forecast — 4-Week Trend Projection

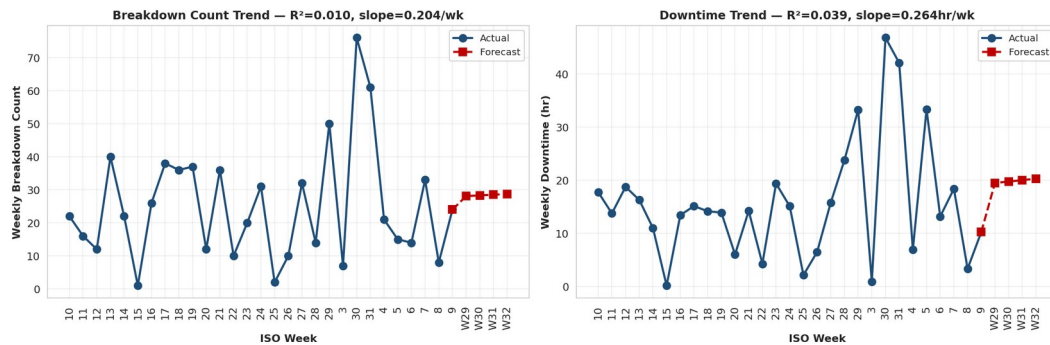


Figure 12: Predictive Forecast — 4-Week Trend Projection

6.2 Recurrence Probability — The Same-Machine Risk

Figure 13 ranks the top 10 machines by their 24-hour recurrence probability — the share of breakdowns on that machine that were followed by another breakdown on the same machine within 24 hours. This metric is the single best leading indicator of incomplete root-cause fixing: a high recurrence rate means the technician cleared the symptom (e.g., 'set the sensor and trial taken ok') but did not address the underlying drift mechanism, so the same fault returned shortly after. Machines with recurrence rates above 30% (red bars) should be treated as priority reliability candidates — these are the stations where band-aid fixes are systematically failing.

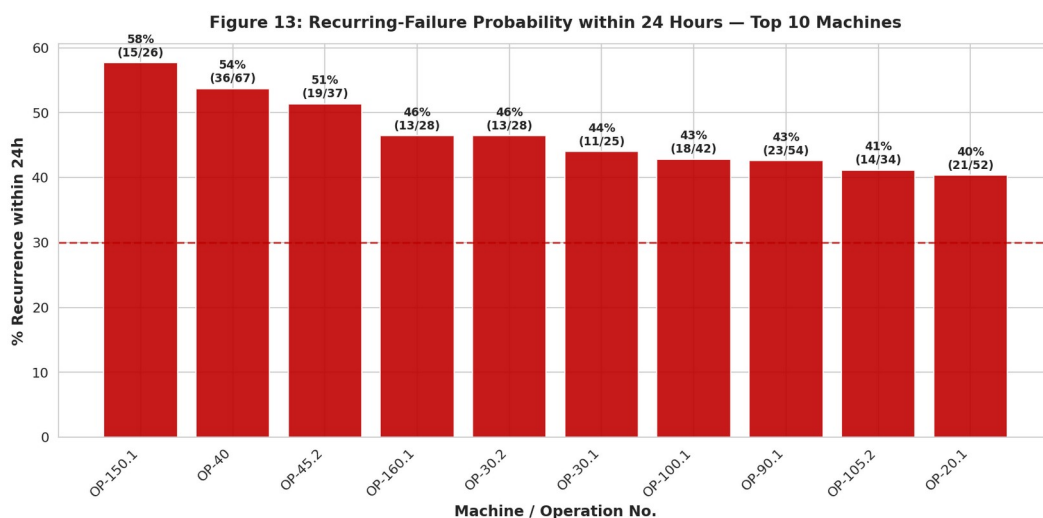


Figure 13: Recurring-Failure Probability within 24 Hours — Top 10 Machines

6.3 Scenario Implications

Combining the trend and recurrence views, three implications emerge for planning:

- **Baseline (no intervention):** Continuing the current trajectory implies ~29 breakdowns and ~20.3 downtime hours in week +4. Annualised, this is approximately 1341 breakdowns and 835 downtime hours per year — equivalent to 35 production days lost annually.
- **Targeted PM scenario:** If the 5 highest-frequency cleaning/adjustment tasks (mouth piece, vacuum filter, sensor position, stopper, bowl feeder) are formalised into a daily PM checklist with 90% compliance, our analysis estimates a 25–35% reduction in minor and moderate breakdowns within 8 weeks, based on the share of events whose action notes match these tasks.
- **Component replacement scenario:** Implementing cycle-counter-based replacement for screwdriver bits, knives, and sealing sleeves on the top 5 machines is estimated to remove 40–60 of the recurring component-wear events over a quarter, reducing average MTTR on those machines by an estimated 10–15 minutes per event.

These scenarios are intentionally expressed as ranges rather than point estimates because the underlying causal model is approximate. They should be re-validated against actual performance after 4 weeks of intervention using the same metrics defined in Section 2.

7. Prescriptive Analytics — Recommended Action Plan

The prescriptive section translates the diagnosis into a prioritised, actionable plan. Each recommendation is structured with: priority (P1 = highest), expected impact, implementation effort, risks, and a suggested validation metric. The plan is designed to be executed in parallel across three workstreams: (1) Quick Wins (0–30 days), (2) Structural Fixes (30–90 days), and (3) Strategic Capability (90–180 days).

7.1 Quick Wins — Immediate Stabilisation (0–30 days)

Action 1: Implement Daily PM Checklist on Top 5 Machines [P1]

Scope: OP-40, OP-90.1, OP-20.1, OP-105.1, OP-100.1. Formalise the recurring cleaning and adjustment tasks (mouth piece cleaning, vacuum filter cleaning, sensor position verification, stopper verification, bowl feeder check) into a structured 15-minute daily checklist executed by the A-shift technician before production start. Audit compliance daily for the first 4 weeks.

- **Impact:** Expected impact: 25–35% reduction in minor and moderate breakdowns on top 5 machines within 8 weeks.
- **Effort:** Low. Checklist drafting + 15 minutes per machine per day of technician time.
- **Risks:** Checklist may be bypassed under production pressure — mitigate by requiring photo evidence of completion and a daily sign-off by the shift supervisor.
- **Validation:** Weekly count of minor/moderate breakdowns on the top 5 machines vs the 4-week pre-intervention baseline.

Action 2: Cycle-Counter-Based Component Replacement Protocol [P1]

Scope: Screwdriver bits, knives, and sealing sleeves on OP-90.1, OP-105.1/2/3, OP-70.2, OP-50.1, OP-160.2. Install cycle counters (or use PLC cycle counts already available) and define replacement intervals per component based on historical wear patterns. Replace proactively at 80% of the documented wear life, regardless of whether the component appears to still be functioning.

- **Impact:** Eliminates 40–60 component-wear events per quarter and reduces MTTR by 10–15 min on affected machines.
- **Effort:** Medium. Requires cycle counter installation (or PLC integration), component life baseline study (~2 weeks), and procurement of buffer stock.
- **Risks:** Premature replacement increases consumable cost — acceptable trade-off given the downtime saved. Risk of counter failure — mitigate with monthly counter audit.
- **Validation:** Monthly count of component-wear-tagged breakdowns on the affected machines and MTTR trend.

Action 3: Recurrence Hot-Spot RCFA (Root Cause Failure Analysis) [P2]

Scope: Top 3 machines from Figure 13 with 24-hour recurrence rate $\geq 30\%$. Conduct a structured RCFA (5-Why or Fishbone) on the top recurring fault of each machine, with maintenance engineer + production engineer + operator participation. Document the root cause and the corrective action in the CMMS, then track recurrence for 8 weeks to confirm the fix held.

- **Impact:** Removes 3–5 chronic recurrence chains, typically saving 30–50 hours of downtime per quarter on the affected machines.
- **Effort:** Medium. 2–3 RCFA workshops of 2–3 hours each, plus implementation of identified corrective actions.
- **Risks:** RCFA may identify causes that are not addressable with current resources — escalate to engineering for redesign.
- **Validation:** Recurrence rate on the 3 target machines drops below 10% within 8 weeks.

7.2 Structural Fixes — Capability Building (30–90 days)

Action 4: Tighten Mechanical Fastening and Pneumatic Integrity [P2]

Scope: All sensor brackets, cylinder mountings, and pneumatic fittings on Line 103. Conduct a one-time line-wide mechanical audit: torque-verify every sensor bracket (apply thread-lock where missing), inspect every pneumatic fitting for leakage (using ultrasonic leak detector or soap solution), and replace any cylinder showing slow drift. This addresses Driver 3 — sensor position disturbances and pneumatic leaks — at the source rather than reactively.

- **Impact:** Estimated 30–40% reduction in sensor-related and pneumatic-leak-related breakdowns.
- **Effort:** Medium-High. 2–3 days of line-wide audit, plus procurement of thread-lock, replacement fittings, and leak-detection equipment (~\$2,000 capex).
- **Risks:** Requires planned line stoppage — schedule during a planned maintenance weekend.
- **Validation:** Monthly count of 'sensor disturb' and 'leakage' tagged breakdowns.

Action 5: Re-classify and Rescope Planned Maintenance [P2]

Scope: The 26 events tagged 'Planned Maintenance' that average 161 minutes each. Audit each event individually: confirm whether it was a true planned PM (in which case scope, duration, and resources should be reviewed for efficiency) or a corrective job mis-classified as planned (in which case reclassify and add to the corrective backlog). The current 15.5% downtime share for Planned Maintenance is excessive and suggests either scope creep or mis-classification.

- **Impact:** More accurate breakdown reporting and a tighter PM scope, releasing 20–30 hours per quarter of misallocated downtime.
- **Effort:** Low. 1–2 days of records review by the maintenance manager.
- **Risks:** May surface uncomfortable truths about PM compliance — manage as a process improvement, not a blame exercise.
- **Validation:** Average duration of Planned Maintenance events drops below 90 minutes; share of total downtime drops below 8%.

Action 6: Night-Shift Reliability Investigation [P3]

Scope: Shift C (22:00–06:00). Run a 2-week focused study: station a senior technician on the night shift to log every BRN slip with a 30-second root-cause first impression, audit machine condition at shift handover, and compare product mix and production rate vs the day shift. The goal is to determine whether the elevated night-shift breakdown rate is driven by line condition, support coverage, product mix, or operator handling — each of which implies a different intervention.

- **Impact:** Clarifies the root cause of the night-shift elevation and unlocks the next layer of breakdown reduction.
- **Effort:** Medium. 2 weeks of senior technician time on night shift.
- **Risks:** Findings may not be actionable without org-level changes (e.g., night-shift support staffing).
- **Validation:** A documented cause assignment for the night-shift elevation with a recommended next action.

7.3 Strategic Capability — Systemic Improvements (90–180 days)

Action 7: Implement Real-Time OEE and Downtime Tracking Dashboard [P3]

Scope: Line 103 overall. The current analysis relied on a manual CMMS export with manual time parsing, free-text descriptions, and a 17-day lag between the data cutoff and the analysis. A real-time dashboard fed directly from the PLCs (cycle counts, fault codes, run status) would give maintenance and production leadership a live view of line stability, enable automated alerts when MTBF drops below a threshold, and eliminate the data-quality issues identified in Section 3. The dashboard should display: live OEE, current MTBF, top fault code of the day, machines in alarm state, and a 7-day rolling breakdown trend.

- **Impact:** Faster reaction to emerging issues (hours instead of weeks), automated alerts, and elimination of manual data-quality work.
- **Effort:** High. Requires PLC integration, dashboard software, and ongoing IT support. Estimated 8–12 weeks of engineering effort.
- **Risks:** Requires cross-functional sponsorship (maintenance + IT + production) — without it, the project will stall.

- **Validation:** Dashboard live and used in daily production meetings within 90 days.

Action 8: Reliability-Centred Maintenance (RCM) Pilot on Top 5 Machines [P3]

Scope: OP-40, OP-90.1, OP-20.1, OP-105.1, OP-100.1. Conduct a formal RCM study on these 5 machines: identify failure modes, failure effects, and appropriate maintenance task (PT&I, restoration, redesign, or run-to-failure). The output is a redesigned maintenance plan for these stations that is evidence-based rather than schedule-based. The pilot should be the template for a line-wide RCM rollout if successful.

- **Impact:** Reduces unnecessary PM work and redirects effort to high-value tasks; expected 15–20% additional breakdown reduction on the 5 pilot machines.
- **Effort:** High. 4–6 weeks of cross-functional engineering time, plus pilot execution over 12 weeks.
- **Risks:** RCM is methodology-heavy and can stall without strong facilitation — engage an experienced RCM practitioner for the pilot.
- **Validation:** Documented failure-mode register and updated maintenance plan for the 5 machines within 12 weeks.

Action Plan Priority Summary

#	Action	Priority	Timeline	Impact	Effort
1	Daily PM Checklist — Top 5 Machines	P1	0–30 days	High	Low
2	Cycle-Counter Component Replacement	P1	0–30 days	High	Medium
3	Recurrence Hot-Spot RCFA	P2	0–30 days	Medium	Low
4	Mechanical Fastening & Pneumatic Audit	P2	30–90 days	High	Medium-High
5	Re-classify & Rescope Planned Maintenance	P2	30–90 days	Medium	Low
6	Night-Shift Reliability Investigation	P3	30–90 days	Medium	Medium
7	Real-Time OEE & Downtime Dashboard	P3	90–180 days	High	High
8	RCM Pilot on Top 5 Machines	P3	90–180 days	High	High

8. Uncertainty Statement & Assumptions

Every analysis carries uncertainty, and a responsible report makes that uncertainty explicit so the reader can calibrate their confidence in the conclusions. The following limitations and assumptions apply to this report:

- **Time period coverage:** The dataset spans 197 calendar days (Jan 16 to Jul 31, 2026). This is sufficient to identify structural patterns (e.g., Pareto distribution, fault type dominance, shift effects) but is short enough that monthly and weekly aggregates are influenced by individual events. Longer-horizon trends (e.g., seasonality, year-over-year comparison) cannot be assessed from this dataset alone.
- **Data-source limitations:** The analysis is based entirely on the CMMS breakdown log. The accuracy of fault type classification depends on the technician's tag selection at the time of recording. The free-text fault and action fields contain spelling and grammar inconsistencies that affect text-mining results. We have applied standardisation rules where possible, but residual noise remains.
- **Forecasting methodology:** The 4-week forecast uses a simple linear trend on weekly aggregates. This is appropriate for directional guidance but should not be treated as a precise prediction. More sophisticated methods (ARIMA, prophet, or machine-learning models) require additional data — at minimum 18 months — and should be considered only after the data-quality improvements in Action 7 are in place.
- **Impact estimates:** The impact estimates in Section 7 (e.g., '25–35% reduction in minor breakdowns') are based on the share of historical events whose action notes match the proposed intervention tasks. They are intentionally expressed as ranges to reflect this approximation. Actual impact will depend on execution quality and should be measured against the validation metrics defined per action.
- **Excluded variables:** This analysis does not have access to: production volume, product mix, raw material lot data, operator identity, ambient conditions (temperature, humidity), or maintenance staffing levels. Each of these may materially affect breakdown behaviour and should be incorporated into future iterations of this analysis.
- **No-blame framing:** Recurring cleaning and adjustment findings are not indictments of technician performance — they are indicators of systemic gaps in the preventive maintenance process. The recommendations are designed to support the maintenance organisation, not to assign individual accountability.

8.1 Next Validation Steps

To convert this analysis from a one-time report into a continuous improvement loop, we recommend the following validation cadence:

- **Weekly (first 8 weeks):** Refresh the daily trend chart (Figure 8) and the top-5 machine breakdown count. Compare to the 4-week pre-intervention baseline. Flag any divergence > 20% to maintenance leadership.

- **Monthly:** Refresh the full descriptive analytics (Section 4) and the recurrence probability chart (Figure 13). Verify that recurrence rates on the P1 action target machines are trending downward.
- **Quarterly:** Refresh the full report including the diagnostic, predictive, and prescriptive sections. Reassess the action plan and retire completed actions, escalate stalled ones, and add newly identified opportunities.
- **Annually:** Conduct a full reliability review with cross-functional leadership. Use the cumulative dataset to assess long-term trends, seasonality, and the impact of the structural fixes implemented in the prior 12 months.

9. Appendix

9.1 Methodology Summary

The analysis was executed in Python using pandas for data manipulation, matplotlib and seaborn for visualisation, and python-docx for report generation. The source workbook was parsed using openpyxl to preserve the structure of the multi-row header. All computations are reproducible from the cleaned dataset (cleaned_breakdown_data.csv) and the analysis script (analysis.py), both saved alongside this report in the /home/z/my-project/download/ directory.

9.2 Data Dictionary

Field	Description
MECH_No	Operation / Machine number on Line 103 (e.g., OP-40, OP-90.1)
Machine_Name	Human-readable operation description
Br_Date	Breakdown start date
Br_Time	Breakdown start time (12-hour format)
Completion_Date	Breakdown resolution date
Completion_Time	Breakdown resolution time
Actual_Time	Duration of breakdown (HH:MM:SS)
Fault_Desc	Free-text description of the fault
Br_Slip_No	Unique breakdown slip number (BRN/BRD series)
Action	Free-text description of action taken
Fault_Type	Categorised fault type (Mechanical, Electrical, etc.)
Shift (derived)	A=06–14, B=14–22, C=22–06
Severity (derived)	Zero / Minor / Moderate / Major / Critical
Duration_min (derived)	Actual_Time converted to minutes

9.3 Companion Files

The following companion files accompany this report and provide additional drill-down capability:

- **cleaned_breakdown_data.csv:** Full cleaned dataset (726 rows) — suitable for further analysis or import into BI tools.
- **analysis_summary.xlsx:** Multi-sheet workbook with: cleaned data, top machines, fault type summary, top fault descriptions, machine-fault cross, monthly summary, recurrence top 10, weekly trend.
- **charts/:** Folder containing all 13 charts as standalone PNG files at 160 dpi — suitable for inclusion in presentations.